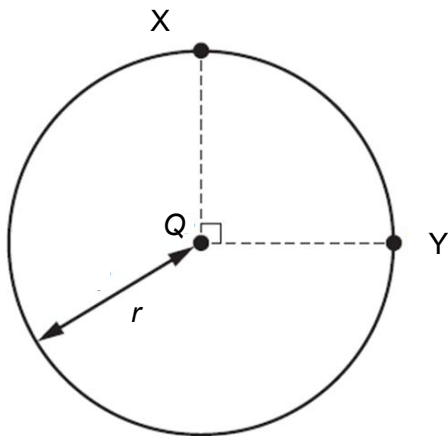


- 19 The diagram shows two points X and Y which lie 90° apart on a circle of radius r .



A positive point charge, Q at the centre of the circle creates an electric field of magnitude $\frac{Q}{4\pi\epsilon_0 r^2}$ at both point X and Y.

Which expression gives the work done in moving a unit positive charge from X to Y?

- A 0 B $\frac{Q}{4\pi\epsilon_0 r}$ C $\frac{Q}{4\pi\epsilon_0 r^2} \left(\frac{\pi r}{2}\right)$ D $\frac{Q}{4\pi\epsilon_0 r^2}$